

1. Administrative & Study Identification

Full Study Title: A Phase 1, open-label, single-dose, multi-center, parallel group study to evaluate the pharmacokinetics of JDQ443 in participants with mild, moderate or severe hepatic impairment compared to matched healthy control participants **Short Title/Acronym:** Phase 1 PK Study of JDQ443 in Hepatic Impairment **Protocol Number:** [Internal Study ID] **NCT Number:** NCT05329623 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** JDQ443 (Opnurasib) **Phase of Development:** Phase 1 **Medical Monitor:** Novartis Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effect of varying degrees of hepatic impairment (mild, moderate, and severe) on the systemic pharmacokinetics (PK), safety, and tolerability of a single oral dose of JDQ443 compared to matched healthy control participants. **Secondary Objectives:** To calculate specific PK parameters including AUCinf, AUClast, and Cmax, and to assess the overall safety and tolerability profile.

2.2 Background & Rationale JDQ443 (opnurasib) is an investigational covalent KRAS G12C inhibitor designed to treat specific solid tumors. Because the liver is a primary organ responsible for processing drugs, hepatic impairment can potentially alter the systemic exposure and safety profile of JDQ443. Evaluating the PK of JDQ443 in participants with varying degrees of impaired liver function is required by health authorities to inform how physicians should prescribe and adjust dosages for this patient population in clinical practice.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Matched Healthy Control Participants) **Blinding:** Open-label **Randomization:** N/A (Parallel group design)

3.2 Planned Enrollment & Duration Target \$N\$: 33 participants (originally planned for up to 48) **Number of Sites:** Multi-center (US) **Estimated Study Start:** May 2022 **Estimated Primary Completion:** April 2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Written informed consent must be obtained before any assessment is performed. 2. Body weight of at least 50.0 kg and Body Mass Index (BMI) within the range of 18 to 40 kg/m². 3. Ability to communicate well with the investigator, comply with study requirements, and willingness to remain confined in the clinical research unit as required by the protocol.

4.2 Exclusion Criteria 1. Contradiction or hypersensitivity to the investigational compound/compound class or its excipients. 2. Pregnant or nursing (lactating) women (confirmed by a positive hCG test). 3. Known history of, or current clinically significant arrhythmias, or a prolonged QTcF interval >480 msec. 4. Use of other investigational drugs within the last 30 days or 5 half-lives prior to dosing, whichever is longer.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: A single dose administration of 200 mg of JDQ443 on Day 1. **Route of Administration:** Oral **Duration of Treatment:** Single-day dosing with subsequent PK sample collection spanning up to Day 4.

5.2 Concomitant & Prohibited Therapy Permitted: Standard stable medical care for underlying conditions that do not interfere with JDQ443 pharmacokinetics. **Prohibited:** Use of other investigational drugs within the last 30 days or 5 half-lives prior to dosing.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -2	Informed Consent, Medical History, BMI verification, Eligibility checks
Baseline	Day -1	Admission to clinical research unit, baseline safety evaluations
Intervention	Day 1	Single 200 mg dose administration of JDQ443, initiation of PK sampling
Follow-up	Days 2 to 4	Continuous blood sample collection for PK characterization, safety monitoring
End of Study	~ Day 30	Post-study safety follow-up contact after last study treatment

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Systemic Pharmacokinetics characterization of JDQ443, specifically determining the Area Under the Curve (AUC_{inf} and AUC_{last}). **Measurement Method:** Blood samples collected from Day 1 through Day 4, with PK variables calculated from plasma concentration-time data using non-compartmental methods.

7.2 Secondary & Exploratory Endpoints - Maximum observed concentration (C_{max}) of JDQ443. - Safety and tolerability profile evaluated by incidence and severity of Adverse Events (AEs), specifically evaluating changes in participants with liver disease vs. healthy participants.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous AE and SAE collection throughout the clinical unit confinement and up to approximately 30 days post-dose during the follow-up contact. **Serious Adverse Events (SAEs):** Evaluated and reported per standard 24-hour reporting protocols. **Data Monitoring Committee (DMC):** Internal Novartis safety and PK review mechanisms.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Analysis Set and Safety Set. **Sample Size Justification:** Approximately 8 participants were targeted for enrollment in each of the mild (Child-Pugh A), moderate (Child-Pugh B), and severe (Child-Pugh C) hepatic impairment groups to ensure at least six evaluable participants per group. Evaluated participants were matched to the healthy control group based on age, body weight, and sex. **Handling of Missing Data:** Standard PK evaluation methodologies; AUC_{inf} and AUC_{last} calculations restricted to participants with evaluable plasma concentration-time profiles.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Continuous monitoring and confinement at the clinical research unit from Day -1 to Day 4. **Audit Trail:** eCRF locking, verifiable PK sample chain of custody, and clinical clarification forms.

11. Study Identifiers & Governance

Primary ID: NCT05329623 **Registry Number:** NCT05329623 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** A Phase 1 Study to Evaluate the Pharmacokinetics of JDQ443 in Participants With Hepatic Impairment Compared to Matched Healthy Control Participants **Official Regulatory Title:** A Phase 1, open-label, single-dose, multi-center, parallel group study to evaluate the pharmacokinetics of JDQ443 in participants with mild, moderate or severe hepatic impairment compared to matched healthy control participants

12. Clinical Framework (Study Specific)

Primary Condition: Hepatic Impairment vs. Healthy Controls
Diagnosis Threshold: Standardized Child-Pugh Score (A, B, C)
Medical Methodology: Pharmacokinetic (PK) Evaluation **Optimization**
Target: JDQ443 dose exposure adjustments for hepatic impairment

13. Study Procedures & Methodology

Management Pathway: Phase 1 clinical pharmacology unit confinement **Education Protocol (HCP):** Site initiation, PK sampling, and matching criteria training **Education Protocol (Patient):** Informed consent, unit confinement expectations, and safety reporting guidance **Quality Audit Mechanism:** Clinical monitoring and rigorous non-compartmental PK blood sample tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to ~58 days per participant (28-day screening + 30-day follow up) **Onsite Visit Cadence:** Confinement from Day -1 through Day 4 **Remote Monitoring Frequency:** Remote safety contact at ~Day 30 **Checklist Review Interval:** Interim safety and PK parameter review prior to enrolling the severe hepatic impairment group

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				